

8.2.37

AI25BTECH11003 - Bhavesh Gaikwad

Question: Find the equation of the conic, that satisfies the given conditions.
vertex $(-3, 0)$, directrix $x + 5 = 0$.

Solution:

Given conditions:

Vertex: $(-3, 0)$

Directrix: $x + 5 = 0$

$\Rightarrow$ Since we have a vertex and directrix, this conic is a parabola with eccentricity $e = 1$.

Using the formulation for a conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e , and focus $\mathbf{F}$:

Given Parameters:

Normal vector: $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Directrix constant: $c = -5$

Eccentricity: $e = 1$

Focus: $\mathbf{F} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ [derived from equidistance property]

Calculating matrix $\mathbf{V}$

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2(\mathbf{n}\mathbf{n}^\top) = \mathbf{I} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (0.1)$$

Calculating $\mathbf{u}$

$$\mathbf{u} = ce^2\mathbf{n} - \|\mathbf{n}\|^2\mathbf{F} = (-5)(1)\begin{pmatrix} 1 \\ 0 \end{pmatrix} - 1\begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} \quad (0.2)$$

Calculating f :

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2e^2 = 1 - 25 = -24 \quad (0.3)$$

The eigenvalues of $\mathbf{V}$ is $\lambda_1 = 0$ (0.4)

Since the eigenvalue is zero, we confirm this is a parabola ($e = 1$).
The general conic equation is:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (0.5)$$

Substituting our values in Equation 0.5:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} -4 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + (-24) = 0 \quad (0.6)$$

$$y^2 - 8x - 24 = 0 \quad (0.7)$$

Therefore, the Equation of the Conic is:

$$\boxed{y^2 = 8(x + 3)} \quad (0.8)$$

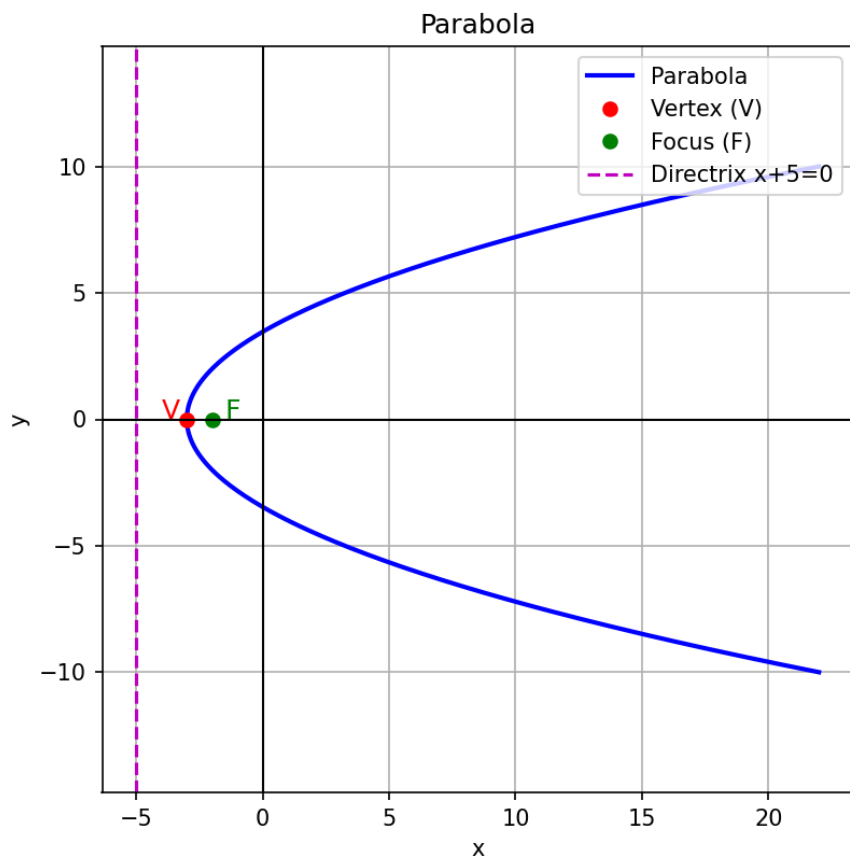


Fig. 0.1: Parabola